Problem-Based Learning

Use the *ASW Example* handout and the *Formula Summary Sheet* (Eqs. (1)–(10), Tables 3.1–3.5, Figs. 3.5–3.6)

Question 1 **Engine choice and take-off weight**

The ASW design example was sized with a **high-bypass turbofan**. Suppose the designers are instead forced to use a **low-bypass turbofan**.

(a) State the new cruise and loiter SFC values.

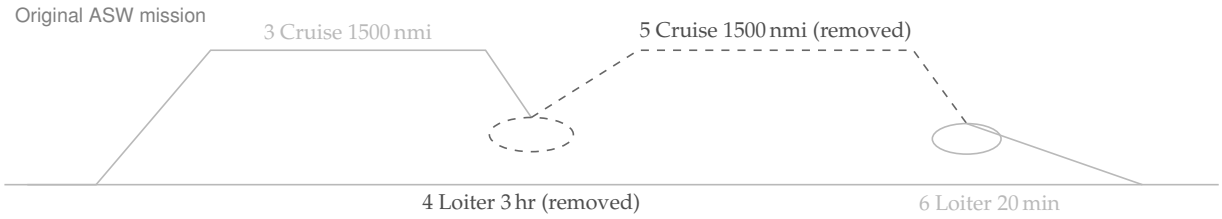
(b) Recalculate the mission segment weight fractions for segments 3–6 and the total mission weight fraction W_7/W_0.

(c) Find the new fuel fraction W_f/W_0 and the sized take-off gross weight W_0.

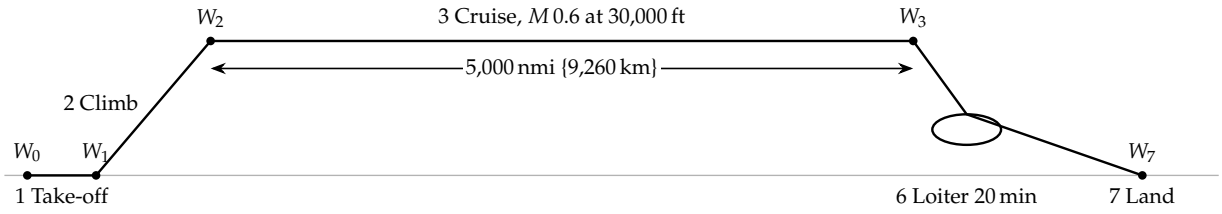
(d) How does the change in SFC affect the take-off gross weight? Comment on why.

Question 2 — Ferry mission

Resize the ASW design example for a **5,000 nmi** {9,260 km} ferry mission: remove segments 4 and 5 (the 3-hour loiter and the return cruise) and make the range of segment 3 equal to the ferry range.



Ferry mission (5,000 nmi)



Question 3 **Advanced engine**

For the ASW design example, what is the effect on range if an advanced engine reduces fuel consumption by **20**%?

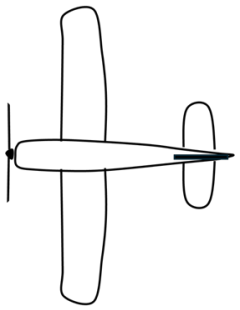
Question 4 **Off-design weight**

For the ASW design example, calculate the total range obtained if the airplane actually weighs **50,000 lb** {22,680 kg}.

Past Exam Question **EMS611U 2025–26, Question 2(a) [16 marks]**

A tractor-configured, four-seat general aviation aircraft suitable for training purposes (see the conceptual sketch).
The aircraft features a **retractable** propeller system, powered by a **piston engine with a fixed-pitch propeller**. The aspect ratio of the aircraft's wing is $A = 8$.

(a) If the aircraft's expected to cruise at **70 m/s** for **1000 km** and loiter for **30 minutes**, calculate the cruise segment mission weight fraction and the loiter segment weight fraction.



Conceptual sketch

You may take gravity as 9.8 N/kg. Useful data on the summary sheet: **Figure 3.6** (S_{wet}/S_{ref}), **Figure 3.5** (L/D_{max} trends) and **Table 3.4** (propeller SFC) — these were Figure 2.2, Figure 2.3 and Table 2.1 in the exam paper.